Adaptive Momentum Memory for Temporal Planning Consistency in End-to-End Autonomous Driving (MemAd)

Project Proposal - Davron M & Alex R

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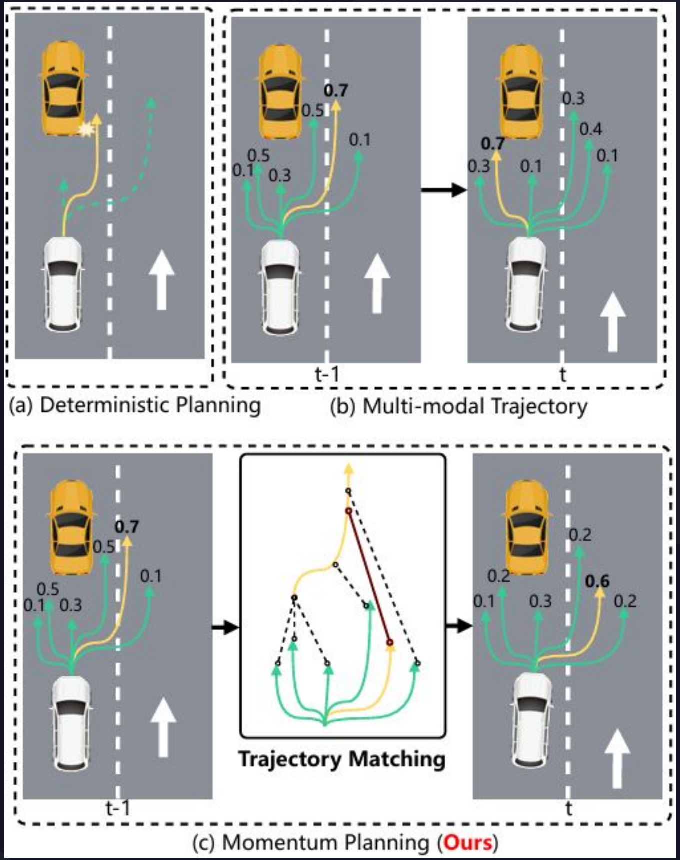
Background

MomAD

- **End-to-End (E2E) Framework:**
  Raw sensor data directly into vehicle motion plans
- **The Problem: Causal Confusion:**
  Past velocity—rather than actual environmental cues
- **Momentum Integration:**
  Temporal consistency in decision-making.



Cont. MomAD - Main metrics

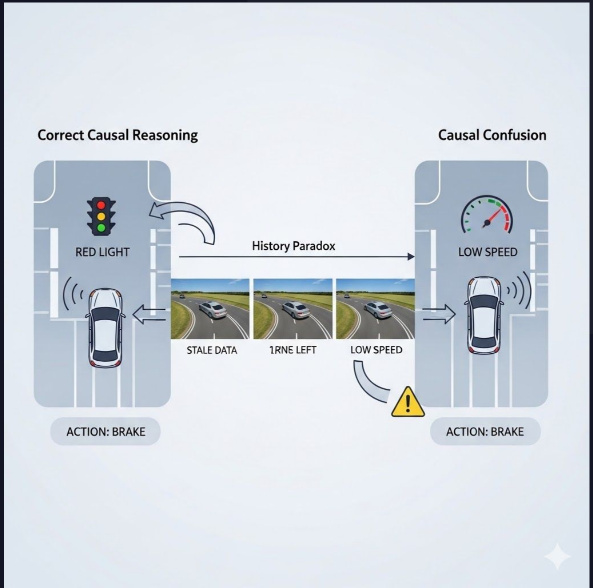
$$\mathrm{TPC} = \frac{1}{N_T}\sqrt{\sum_{i=1}^{N}\left(\left(T_t^{\mathrm{Pred}} - T_{t-1}^{\mathrm{Pred}}\right)^2 \cdot T_{\mathrm{GT}}^{\mathrm{Mask}}\right)}, \quad (6)$$

where N_T is the total number of GT trajectories in the validation set, and $T_{\mathrm{GT}}^{\mathrm{Mask}}$ is the mask for trajectories exceeding the overlapped time period of two trajectories. Our TPC metric evaluates whether autonomous vehicles adhere to predicted trajectories, ensuring continuity across frames. Notably, the TPC metric provides a statistical perspective on the dataset-wide evaluation rather than at the individual sample level.

- **L2 Displacement Error:**
  How far a predicted trajectory is from the ground-truth trajectory. Avg. Euclidian Distance
- **Collision Rate Benchmark:**
  How often the predicted ego trajectory intersects with other agents or map obstacles.
- **Key Metric: TPC**
  Trajectory Prediction Consistency

Current challenges

- **The "History Paradox":** Ablation studies show that increasing history length (t) does not linearly improve performance; specifically, t=2 outperformed t=3, indicating that more history can degrade planning quality.
- **Causal Confusion (The Copycat Problem):** Models frequently latch onto "spurious correlations," such as current velocity or past trajectory, instead of reacting to immediate visual cues like traffic lights.



Adaptive History Length for Momentum Planning

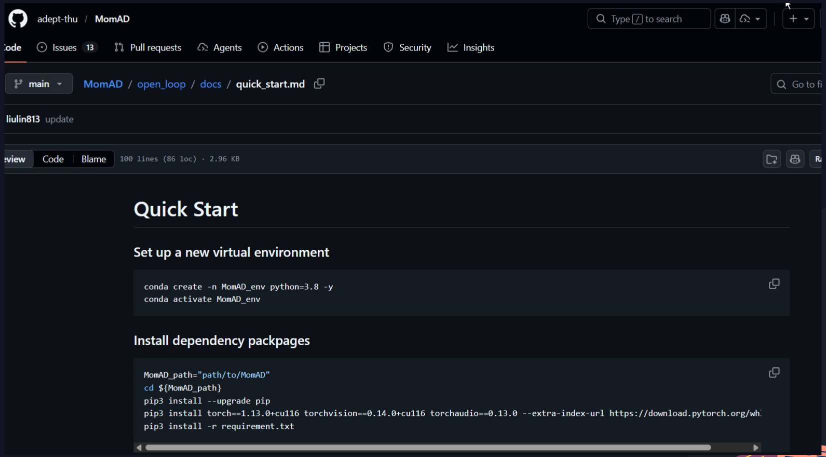
- Limitation
  - MPI uses a fixed amount of historical information, even though the best history length may vary by scene.
- Hypothesis
  - The optimal planning history should depend on scene context and planning uncertainty.
- Improvement
  - Adaptive history selector for MPI that chooses whether to use: no history, 1 previous frame, 2 previous frames

Table 7. Impact of history frames in MomAD on the Turning-nuScenes validation set. t denotes the frame number, where 1 indicates the history is empty (represented by 0), 2 signifies that the historical result corresponds to the previous 1 frame, and 3 indicates that the historical result pertains to the previous 2 frames. We follow the VAD [19] evaluation metric.

ED	MP	NS	t	L2(m) ↓			Col. Rate(%) ↓			TPC(m) ↓		
				2s	3s	Avg.	2s	3s	Avg.	2s	3s	Avg.
				0.77	1.46	0.86	0.17	0.98	0.40	0.70	1.33	0.79
	✓		2	0.71	1.27	0.77	0.14	0.83	0.34	0.55	1.07	0.65
✓		0.1		0.77	1.45	0.86	0.18	0.97	0.39	0.69	1.33	0.79
✓	✓	0.1	1	0.78	1.48	0.88	0.18	0.98	0.39	0.70	1.35	0.81
✓	✓	0.1	2	**0.70**	**1.24**	**0.76**	**0.13**	**0.79**	**0.32**	**0.54**	**1.05**	**0.63**
✓	✓	0.1	3	0.72	1.27	0.78	0.14	0.84	0.35	0.56	1.09	0.66

Steps Taken

1. Reproduce official MomAD baseline
2. Keep TTM, denoising, and loss definitions unchanged
3. Modify only the MPI temporal-history pathway
4. Verify performance of history usage being adaptive per scene instead of fixed



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Improvement

Previous MPI Pipeline

- Previous planning score
- Previous planning query
- Current selected planning query

Long Horizon Query Mixer always tries to enrich the current query with the available past planning state

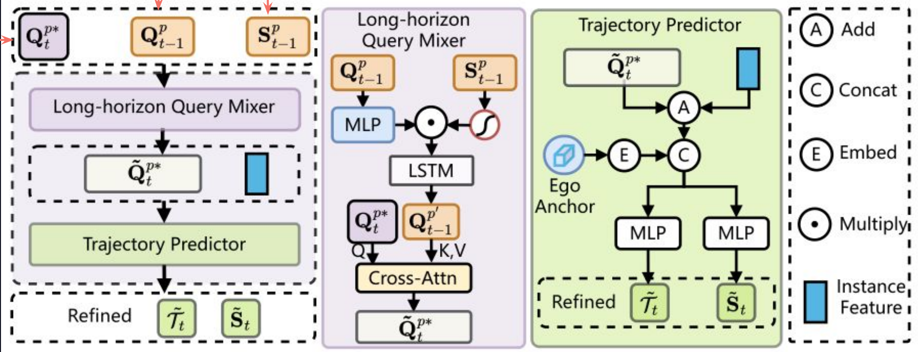
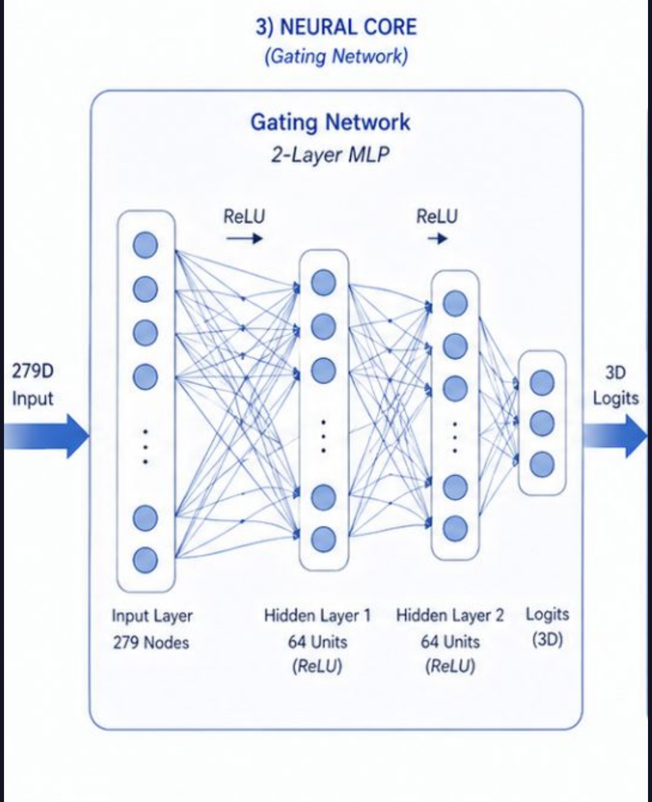


Figure 3. **The illustration of Momentum Planning Interactor (MPI).** MPI cross-attends a selected planning query with historical queries to expand static and dynamic perception files, resulting in an enriched query that improves long-horizon trajectory generation and reduces collision risks.

Improvements in MLP



MLP (Multi-Layer Perceptron) is a fundamental type of artificial neural network

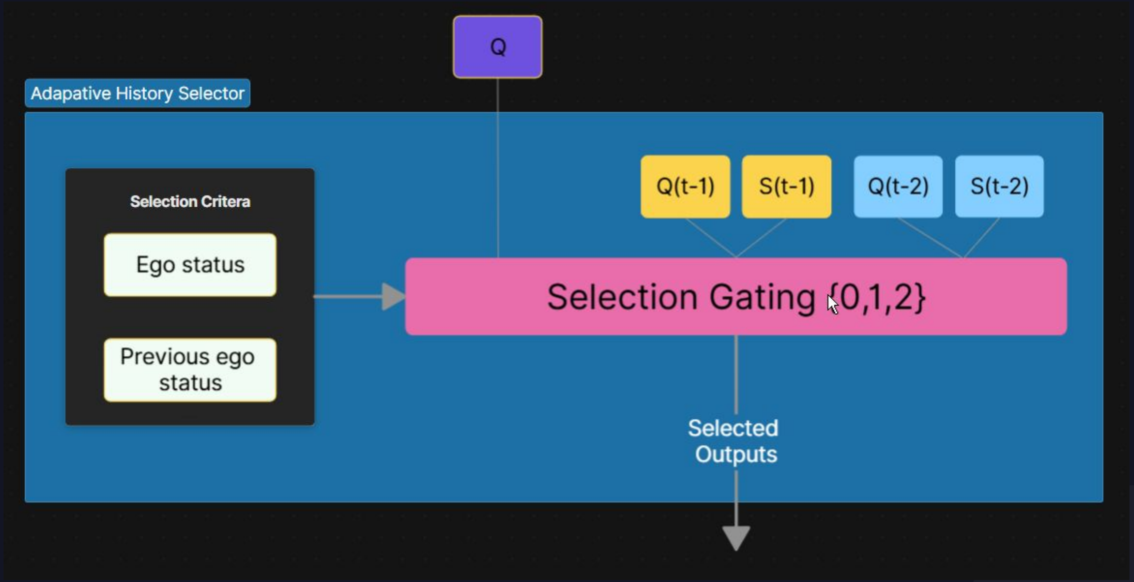
How the MLP Operates in MemAd:

- The Gating Brain
- Technical Architecture
- Optimized Learning

Adaptive History Selector

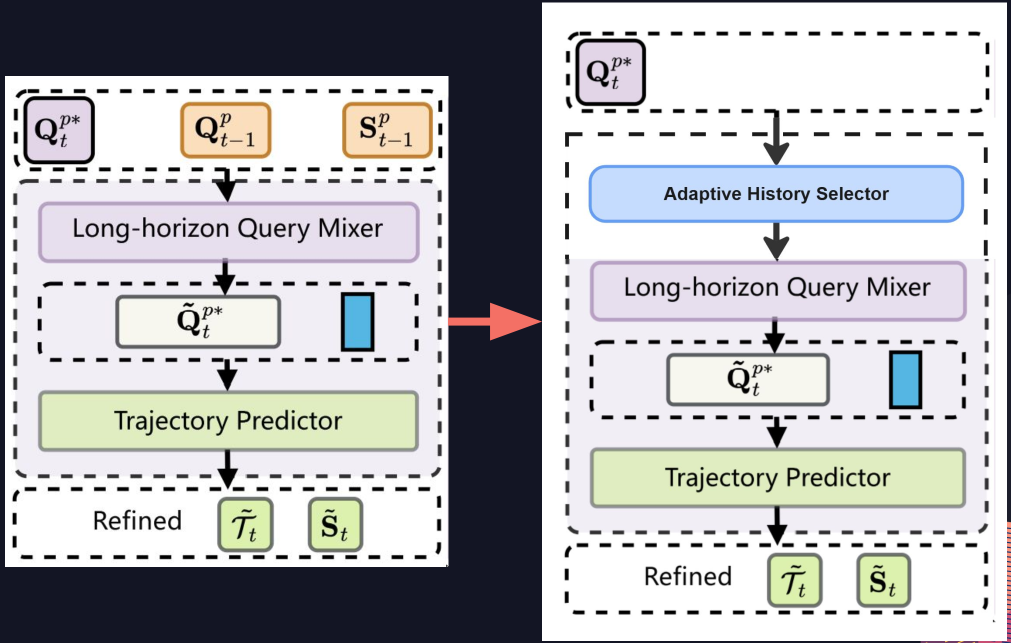
Ego (car) status data

- Steering angle
- Rotation rate
- Velocity
- Acceleration



MPI Pipeline

- New module: Adaptive History Selector
  - Insert a lightweight selector before the Long-horizon Query Mixer
- Selector predicts history depth
  - 0, 1, or 2



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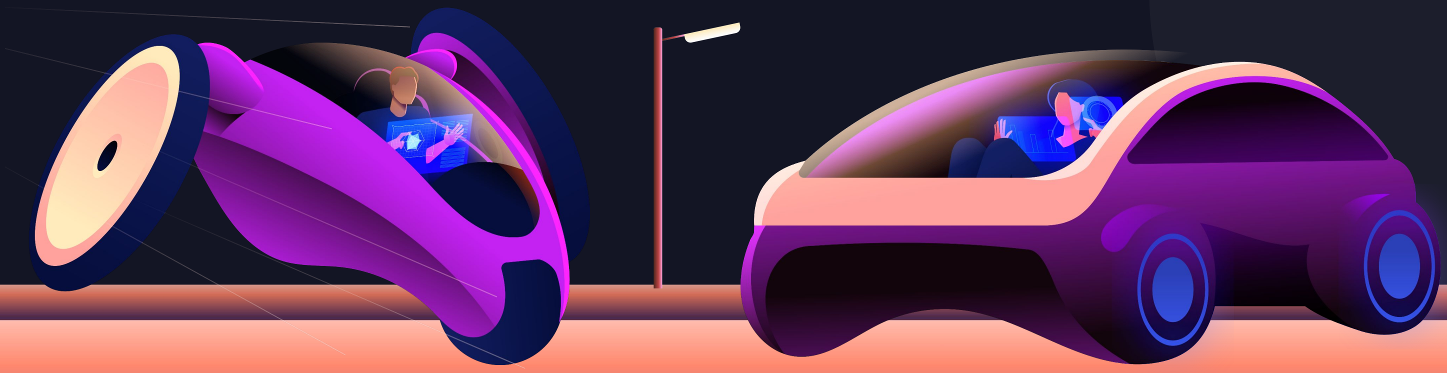
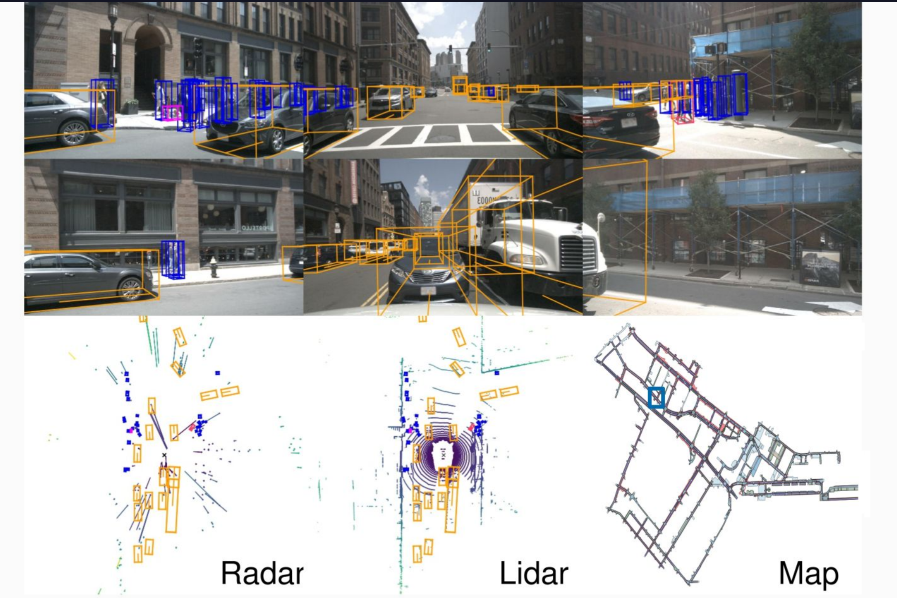
Experiment



Setup

- UA HPC environment
  - 1 P100 GPU (Ocelote)
- Mini dataset (323 / 81 Samples)
  - 35GB (HPC has 50GB limit per user)
  - Full dataset is 400GB (6000 / 150)
- Lacked support for FlashAttention
  - Needs compute 8
  - P100 is compute 7



"Ped with pet, bicycle, car makes a u-turn, lane change, peds crossing crosswalk"

Baseline

- Limited dataset → Numbers are worse than the baseline
- Use limited MomAD as our baseline for improvements

Lower is better for all metrics except FPS*

Method	L2 Avg	Col Avg	TPC	FPS
UniAD	0.73	0.61	0.68	1.8 (A100)
SparseDrive	0.61	0.08	0.57	9.0 (4090)
MomAD	0.60	0.09	0.54	7.8 (4090)
Mini MomAD*	1.44	0.32	0.88	1.4 (P100)
Degradation	2.4X	3.55X	1.63X	5.57X

MemAD 1.0

- MemAD 1.0 (22K Params)
  - Minimum viable implementation of our softgate
- Dataset is small, improvements are within noise
  - P < 0.10 for a trend
  - P < 0.05 significant

Lower is better for all metrics except FPS*

Method	L2 Avg	Col Avg	TPC	FPS
Mini MomAD*	1.44	0.32	0.88	1.4 (P100)
MemAD 1.0	1.39	0.32	0.95	1.4 (P100)
1.0 Improvement	3.5%	0%	-7.9%	0%
P-value	0.41	1.00	0.27	

MemAD 2.0

- MemAD 2.0
  - Added “LEFT/RIGHT/STRAIGHT” input into MLP
  - Warm-start initialization
  - Entropy regularization
- L2 metric trends in the right way
- One less collision
- TPC (consistency) trends in the wrong way

Lower is better for all metrics

Method	L2 Avg	Col Avg	TPC
Mini MomAD*	1.44	0.32	0.88
MemAD 1.0	1.39	0.32	0.95
1.0 Improvement	3.5%	0%	-7.9%
1.0 P-value	0.41	1.00	0.27
MemAD 2.0	1.33	0.16	0.99
2.0 Improvement	7.6%	50%	-12.5%
2.0 P-value	0.067	0.72	0.067

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Discussion

Workflow explanation

1. The Decision Maker:
   adaptive_history_selector.py

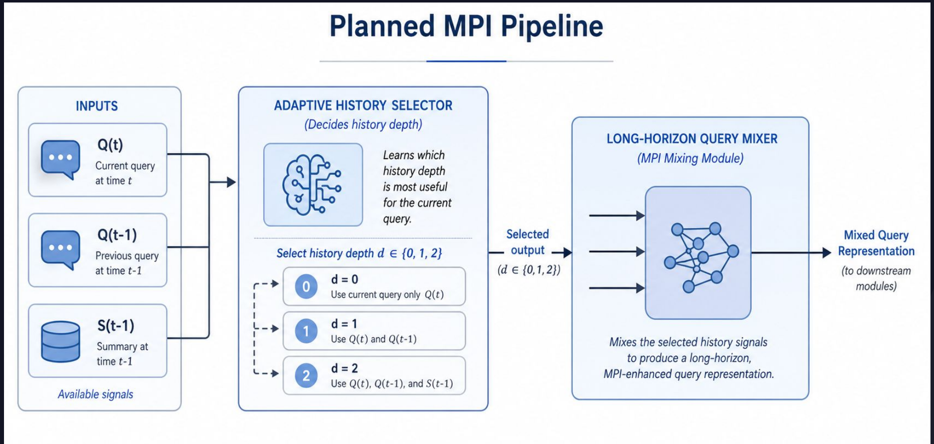
   A. The Input (279 Dimensions)
   B. Softmax Output
   C. The "Warm-Start" Trick

2. The Memory Combiner:
   next_token_prediction.py

   A. The ConsistencyLSTM
   B. The NextTokenPredictor

3. The Central Hub:
   motion_planning_head_roboAD.py

   A. State Caching
   B. Dynamic Fusion
   C. Entropy Regularization



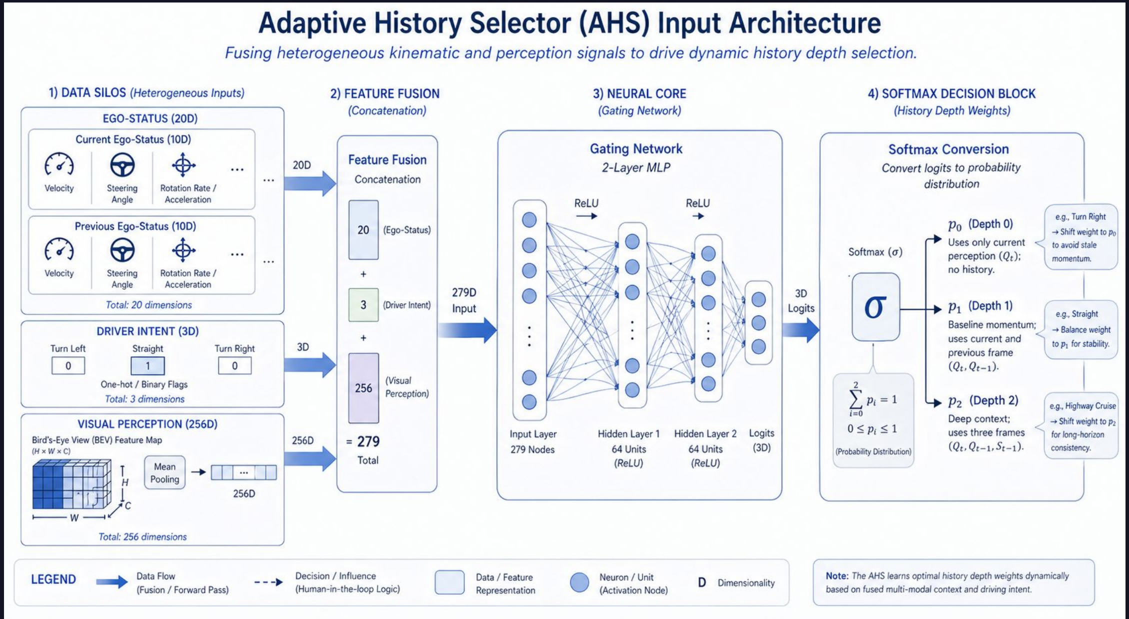
Workflow Explanation

4. The Training Catalyst: MomAD_small_stage2_roboAD.py

   A. Targeted Learning Rates

5. The Stabilizers: planning_eval_roboAD.py & attention.py

   A. The NaN-Guard (Eval)
   B. The Fallback Fix (Attention)



Gate Behavior: Did it Learn?

- Gate strongly prefers branch 1, just like the paper

Branch	What it routes to	% Chosen
0	No history	4.1%
1	T-1 (original MPI)	88.3%
2	T-1 + T-2	7.6%

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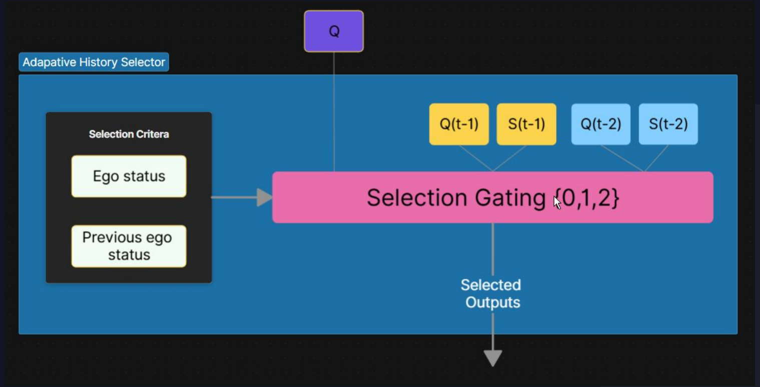

Bimodal Turn Theory

- Bimodal turn behavior theory
  - Some turns are sharp and some are soft
  - Sharp turns, better to have no history
  - Long turns, history is extremely important

Command	n	Branch 0	Branch 1	Branch 2
TURN_RIGHT	132	9.8%	78.0%	12.1%
TURN_LEFT	42	0.0%	88.1%	11.9%
STRAIGHT	312	2.2%	92.6%	5.1%

Conclusion

1. Resolution of the History Paradox
2. Quantitative Performance Gains
3. Behavioral Policy Discovery
4. Methodological Validation

Next Steps

- Train on full dataset
  - More disk space
  - More and better GPUs
- Investigate bimodality turn theory
  - Hyper parameter testing
- Closed-loop testing

Acknowledgements

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Acronyms:

TPC - Trajectory Prediction Consistency

MPI - Momentum Planning Interactor

TTM - Topological Trajectory Matching

Thanks!

Do you have any questions?